



Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories — spending patterns, customer segments, and subscription behavior to guide strategic decisions.

Dataset at a Glance

3,900

Transactions

Across all product categories

18

Features

Demographics, purchases & behavior

37

Missing Values

In Review Rating column only

50

Locations

Geographic coverage

Key features include customer demographics (age, gender, location, subscription), purchase details (item, category, amount, season), and shopping behavior (discounts, promo codes, frequency, review ratings, shipping type).

Data Preparation in Python

01

Load & Explore

Imported via `pandas`; used `df.info()` and `.describe()` for structure and summary stats

02

Clean & Standardize

Imputed missing Review Ratings using median per category; renamed columns to `snake_case`

03

Feature Engineering

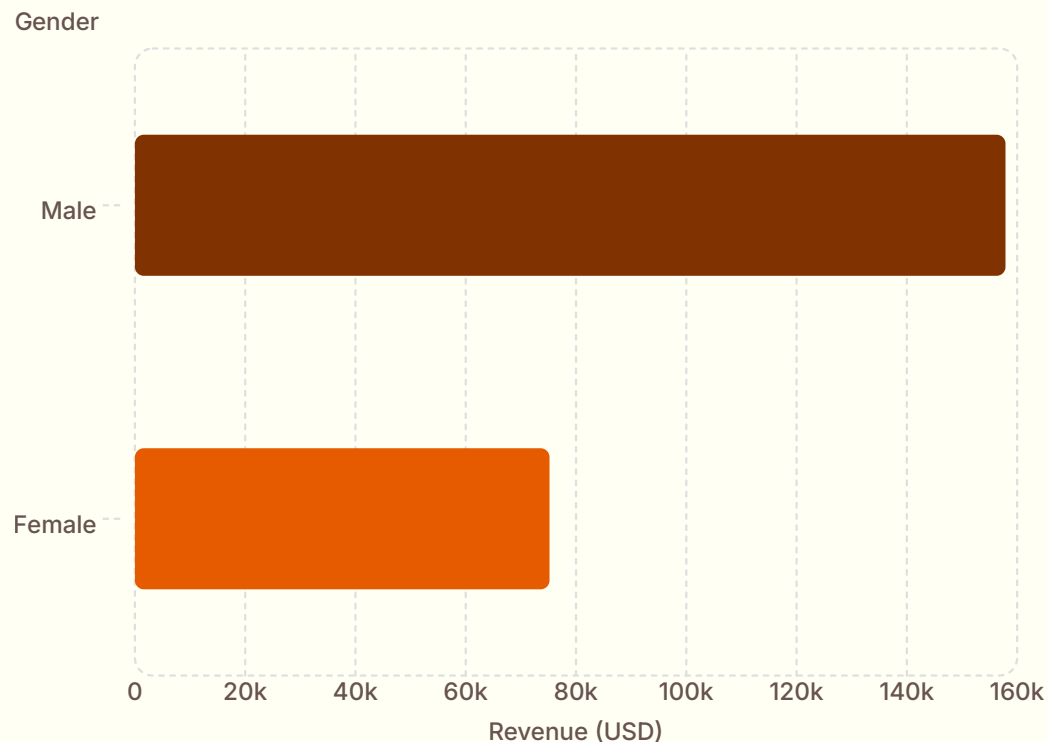
Created `age_group` bins and `purchase_frequency_days`; dropped redundant `promo_code_used`

04

Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis

Revenue by Gender



Key Finding

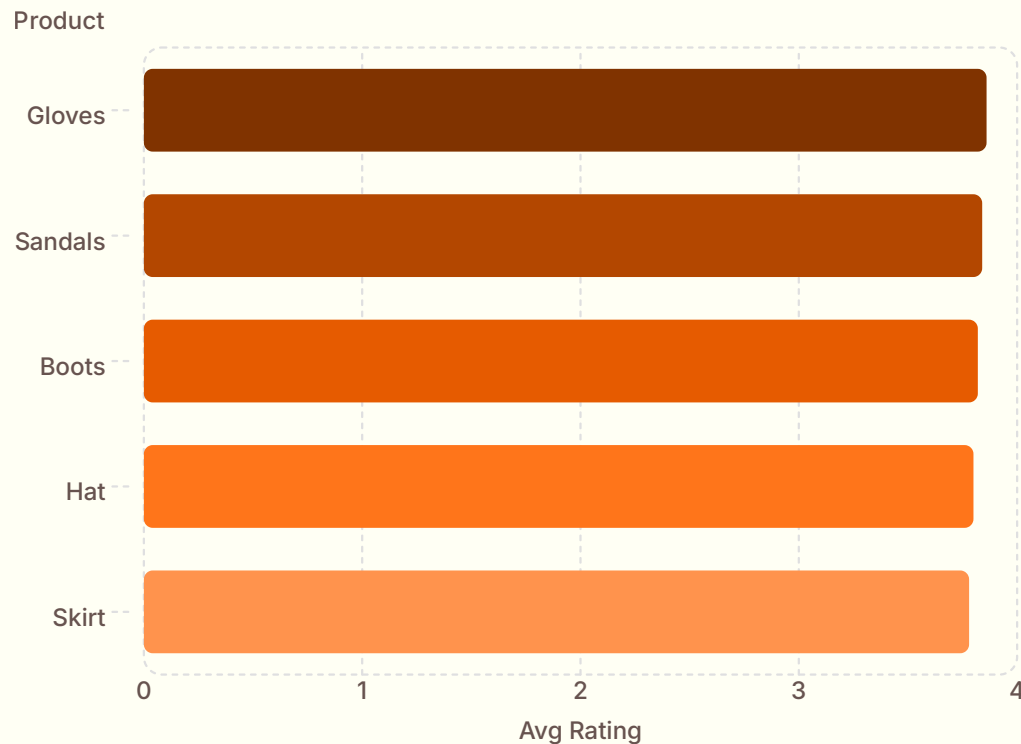
Male customers generated **\$157,890** in revenue — more than double the **\$75,191** from female customers.

Despite females making up the majority of records (2,652 vs. 1,248 inferred), male shoppers spend significantly more per transaction on average.

i This gap presents a clear opportunity to increase female customer engagement and average order value.

Product Insights

Top 5 Products by Rating



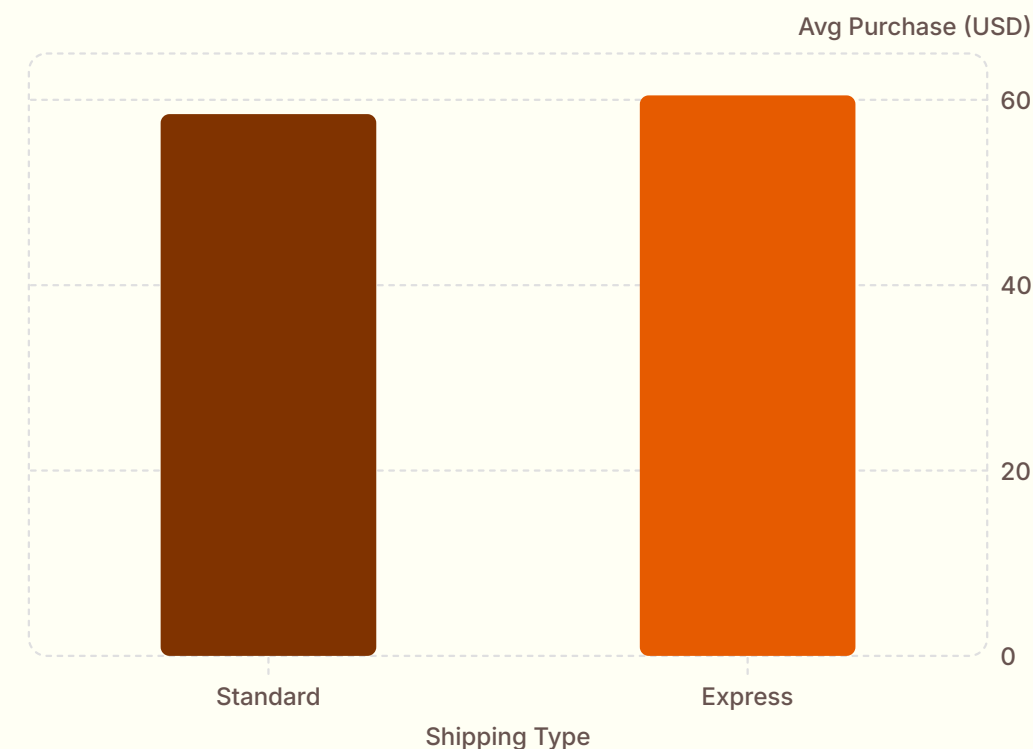
Most Discount-Dependent Products

- **Hat** — 50.00% of purchases discounted
- **Sneakers** — 49.66%
- **Coat** — 49.07%
- **Sweater** — 48.17%
- **Pants** — 47.37%

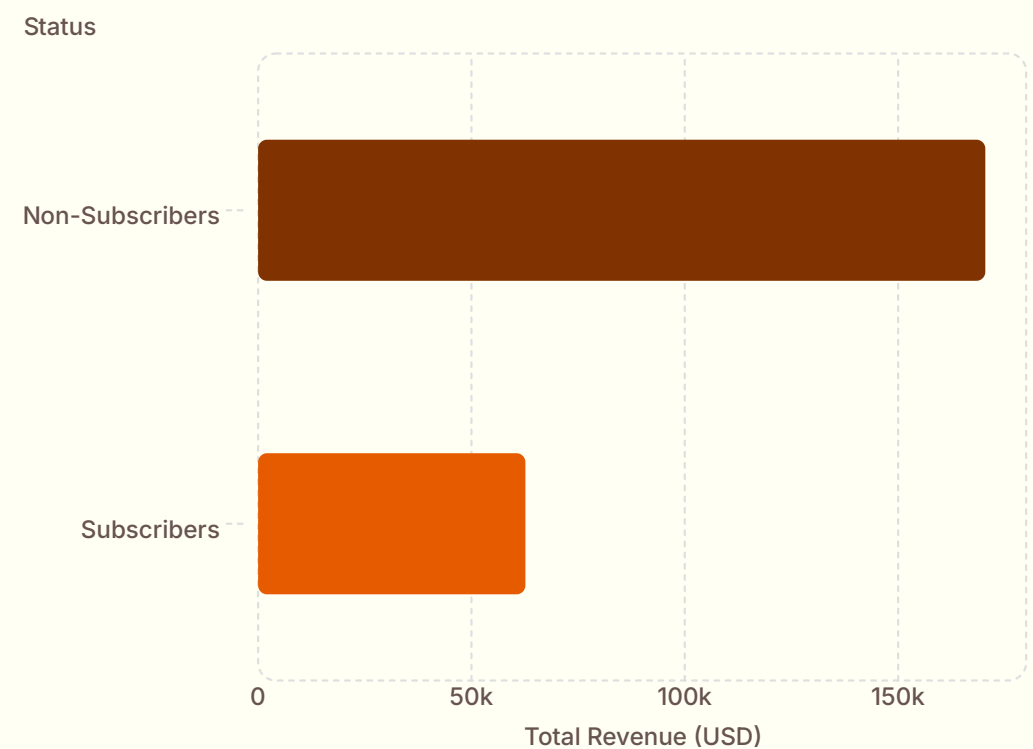
These five products rely most heavily on discounts to drive purchases — a key consideration for margin management.

Shipping & Subscription Analysis

Avg Purchase by Shipping Type

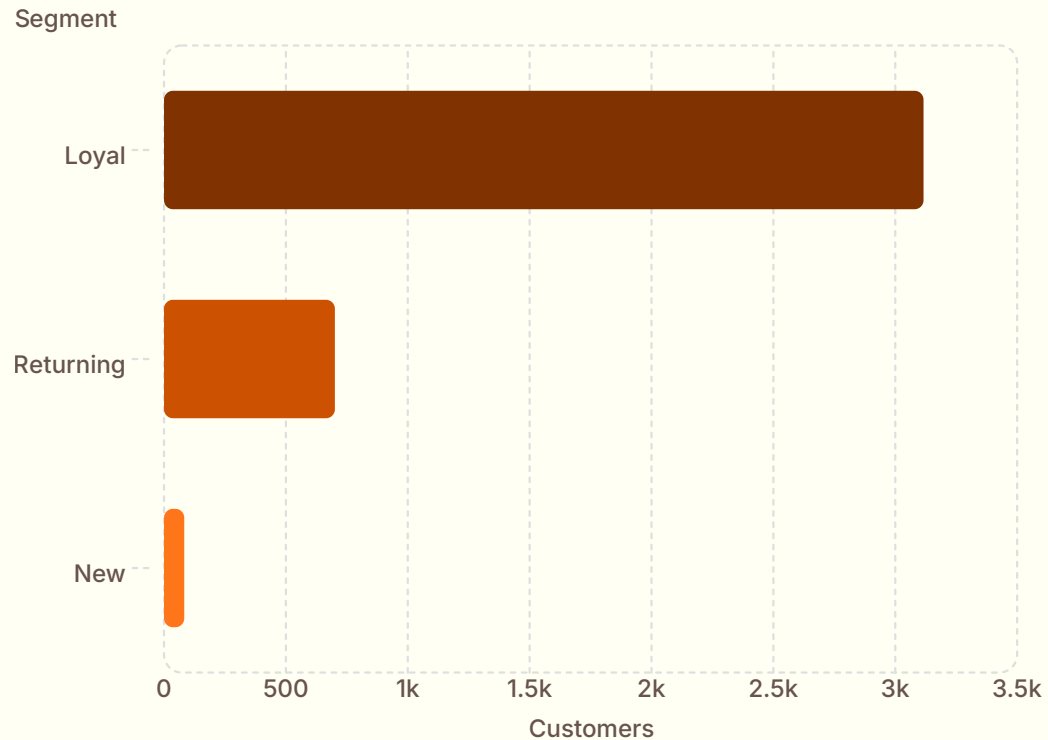


Subscribers vs. Non-Subscribers



Express shipping users spend slightly more on average (**\$60.48** vs. **\$58.46**). Non-subscribers drive **\$170,436** in revenue vs. **\$62,645** for subscribers — a major growth opportunity.

Customer Segmentation



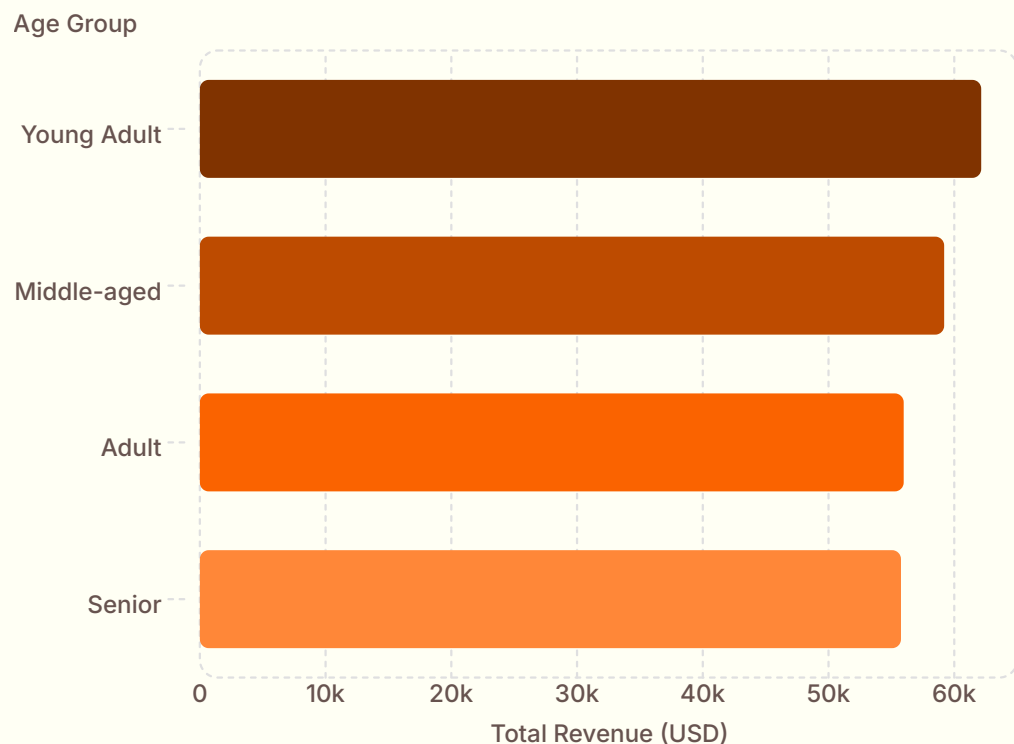
Segment Breakdown

Customers classified by purchase history into three tiers:

- **Loyal (3,116)** — The dominant segment, representing the core revenue base
- **Returning (701)** — Active shoppers with growth potential
- **New (83)** — Smallest segment, key target for onboarding campaigns

Of customers with 5+ purchases, 2,518 are non-subscribers vs. 958 subscribers — a clear upsell opportunity.

Revenue by Age Group



Key Insight

Young Adults lead revenue at **\$62,143**, with other age groups closely clustered between **\$55,763–\$59,197**.

Revenue distribution is relatively even across all four age groups, suggesting broad product appeal. Targeted marketing should focus on Young Adults while maintaining engagement across all segments.

Power BI Dashboard

KPIs

3.9K customers · \$59.76 avg purchase · 3.75 avg rating

Filters

Subscription, Gender, Category, Shipping Type

Visualizations

Revenue & sales by category and age group; subscription donut (27% Yes, 73% No)



Business Recommendations

→ Boost Subscriptions

Promote exclusive subscriber benefits to convert the 73% non-subscriber base

→ Customer Loyalty Programs

Reward repeat buyers to advance them into the "Loyal" segment

→ Review Discount Policy

Balance sales boosts with margin control on discount-dependent products

→ Product Positioning

Highlight top-rated and best-selling products in marketing campaigns

→ Targeted Marketing

Focus on high-revenue age groups and express-shipping users